11.7 Homework: Calculus Review — Markscheme

— Section A: No Calculator —

1. [Maximum mark: 4]

   $g(x) = \mathrm{e}^{x^2+1}$

   $g'(x) = 2x \cdot \mathrm{e}^{x^2+1}$ **M1A1**

   $g'(-1) = 2(-1) \cdot \mathrm{e}^{(-1)^2+1}$ **M1**

   $g'(-1) = -2\mathrm{e}^2$ **A1**

2. [Maximum mark: 5]

   (a) $\dfrac{3\sqrt{x}-5}{\sqrt{x}} = 3 - 5x^{-1/2}$

   $p = -\dfrac{1}{2}$ **A1**

   (b) $\displaystyle\int_1^9 \left(3 - 5x^{-1/2}\right) \mathrm{d}x$

   $= \left[3x - 10x^{1/2}\right]_1^9$ **M1A1A1**

   $= (27-30) - (3-10)$

   $= -3 + 7 = 4$ **A1**

3. [Maximum mark: 5]

   $y = (2x-1)\mathrm{e}^{kx}$

   $y' = 2\mathrm{e}^{kx} + (2x-1) \cdot k\mathrm{e}^{kx}$ **M1A1**

   At $x = 1$: $y' = 2\mathrm{e}^k + k\mathrm{e}^k = (2+k)\mathrm{e}^k$ **A1**

   Tangent parallel to $y = 5\mathrm{e}^k x$, so slope $= 5\mathrm{e}^k$:

   $(2+k)\mathrm{e}^k = 5\mathrm{e}^k$ **M1**

   $k = 3$ **A1**

4. [Maximum mark: 6]

   (a) Perimeter of shaded region $= 2r + r\theta = 2r + 2r = 4r$ **M1A1**

   $4r = 20$

   $r = 5$ **A1**

(b) Shaded area $= \frac{1}{2}r^2\theta = \frac{1}{2}(25)(2) = 25$ **M1**

Total area $= \pi r^2 = 25\pi$ **A1**

Non-shaded area $= 25\pi - 25$ **A1**

5. [Maximum mark: 7]

(a) $f'(4) = 6$ **A1**

(b) $f(4) = 6(4) - 1 = 23$ **A1**

(c) $g(4) = 16 - 12 = 4$ **M1**

$h(4) = f(g(4)) = f(4) = 23$ **A1**

(d) $h'(x) = f'(g(x)) \cdot g'(x)$ **M1**

$g'(x) = 2x - 3$, so $g'(4) = 5$

$h'(4) = f'(4) \cdot 5 = 6 \times 5 = 30$ **A1**

Tangent: $y - 23 = 30(x - 4)$

$y = 30x - 97$ **A1**

— Section B: Calculator Allowed —

6. [Maximum mark: 6]

 (a) $\int (6x+7)\,\mathrm{d}x = 3x^2 + 7x + C$ **M1A1A1**

 (b) $f(x) = 3x^2 + 7x + C$

 $f(1.2) = 3(1.44) + 7(1.2) + C = 12.72 + C = 7.32$ **M1A1**

 $C = -5.4$

 $f(x) = 3x^2 + 7x - 5.4$ **A1**

7. [Maximum mark: 5]

 $g'(x) = 3x^2 + 5\mathrm{e}^x$

 $g(x) = x^3 + 5\mathrm{e}^x + C$ **M1A1A1**

 $g(0) = 0 + 5 + C = 4$ **M1**

 $C = -1$

 $g(x) = x^3 + 5\mathrm{e}^x - 1$ **A1**

8. [Maximum mark: 5]

 (a) Using GDC: $a \approx 8.57$, $b \approx -36.9$ **A1A1**

 (b) $r \approx 0.995$ **A1**

 (c) $y = 8.57(30) - 36.9 = 220.2$ **M1**

 ≈ 220 drinks **A1**

9. [Maximum mark: 5]

 (a) Sketch: $f(0) = -1$, curve rises to maximum near $(1.74, 4.4)$, then falls through zero and decreases to $f(3) \approx -1.3$.

 Correct shape with maximum **A1**

 Correct endpoints and intercepts **A1A1**

 (b) $f'(x) = 0$ at $x \approx 1.74$ (GDC) **M1A1**

10. [Maximum mark: 6]

 $f(x) = \ln(x\mathrm{e}^x + 1) - x^4$

 (a) Using GDC: A $\approx (0.720, 0.639)$ **A1A1**

(b) $x_{\mathrm{B}} \approx 1.10$ **A1**

(c) Total area $= \displaystyle\int_0^{1.10} f(x)\,\mathrm{d}x - \int_{1.10}^{2} f(x)\,\mathrm{d}x$ **M1M1**

$\approx 0.289 + 13.9 \approx 14.2$ **A1**